



IIIT Bangalore

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Question 59

Question

Two push buttons P_1 and P_2 control the display of characters on a seven segment display.

Determine the logic expression for segment g .

Analysis

The truth table for inputs P_1 and P_2 is

P_1	P_2	g
0	0	0
1	0	1
0	1	1
1	1	1

From the table we observe that the segment turns ON when either P_1 or P_2 is 1.

Therefore the Boolean expression is

$$g = P_1 + P_2$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

P1 = np.array([0,1,0,1])
P2 = np.array([0,0,1,1])

g = P1 | P2

print("P1 P2 g")

for i in range(4):
```

```

print(P1[i], P2[i], g[i])

x = np.arange(4)

plt.step(x,P1,where='post',label='P1')
plt.step(x,P2,where='post',label='P2')
plt.step(x,g,where='post',label='g')

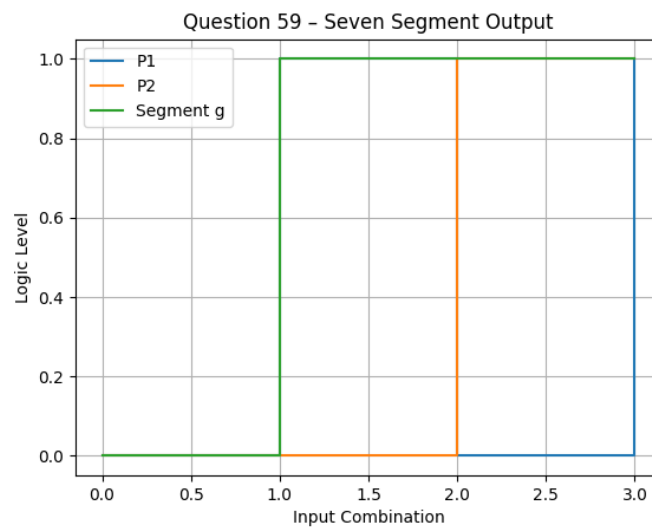
plt.grid()
plt.legend()

plt.savefig("graph.png")

plt.show()

```

Graph



Conclusion

The output follows the Boolean expression $g = P_1 + P_2$. The segment becomes active whenever either of the push buttons is pressed. The Python graph confirms the OR gate behavior.